Architecture Selection – Patient Health Record Consent Management System

Architectural Choice

We chose Layered Architecture for the Patient Health Record Consent Management System.

Reason 1 – Separation of Concerns

The system separates the **Presentation Layer, Business Logic Layer, and Data Layer**. The Presentation Layer handles user interaction, while the Business Logic Layer manages authentication, patient records, consent management, and consent validation. The Data Layer stores patient health records and consent-related information. This separation makes the system easier to understand, maintain, and modify.

Reason 2 – Suitable for the System

The Patient Health Record Consent Management System contains closely related functions such as authentication, patient record management, consent management, and controlled access. Layered Architecture allows these functions to be organized into clear components and enables communication between the user interface, business logic, and database through defined interfaces.

Security Advantage

Layered Architecture allows security controls to be handled within the Business Logic Layer. The **Authentication** and **Access Control / Consent Validation** components can verify the user's identity and consent permissions before allowing access to patient health records. This helps prevent unauthorized access to sensitive patient information.

Performance Benefit

The layered structure keeps the main business operations within the application and provides direct interaction with the data layer. This can reduce unnecessary communication overhead and support faster retrieval of patient records under normal operating conditions.